Melting & Boiling

Exam code: 0625 & 0972

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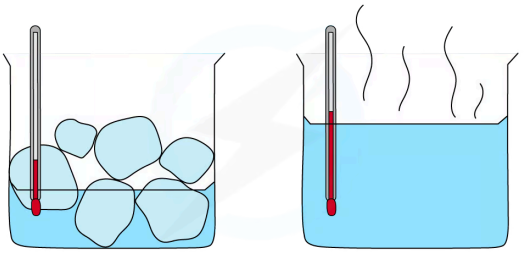
Fixed points of water

- The melting and boiling points of pure water are known as **fixed points**
  - Ice melts at 0 °C
  - Pure water boils at 100 °C
- These are the accepted values for **pure water** at **atmospheric pressure**



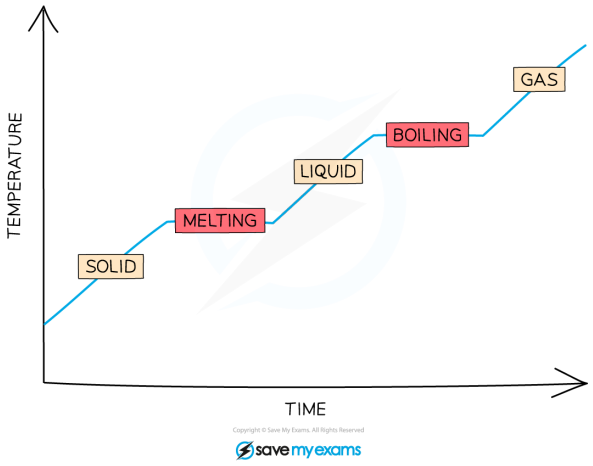
Ice melts at 0 °C and water boils at 100 °C

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Melting & boiling

the thermal energy store of the substance
- One major difference between melting and boiling is that boiling occurs at higher energy
  - While melting results in particles being able to flow and move more freely, during boiling enough energy is transferred such that the intermolecular forces can be completely **overcome**



When the substance is changing state, the temperature remains constant as shown by the horizontal regions of this graph

Boiling

- When liquid water is heated by adding thermal energy (say from a gas flame or kettle element), the temperature of the water rises until the water boils
  - At the **boiling** point, even if more thermal energy is added, the temperature of the liquid water does not increase
  - This means that the internal energy is not rising
- The additional thermal energy goes into overcoming the intermolecular forces between the molecules of water
  - As the forces are overcome, the liquid water becomes water vapour (steam)
  - This is **evaporation** or vaporisation; the water is now a **gas**

Melting

- When solid water (ice) is heated by adding thermal energy, the temperature of the ice increases up until the melting point
  - At the **melting** point, even if more thermal energy is added, the ice does not get warmer
  - This means that the internal energy is not rising
- The additional thermal energy goes into overcoming the intermolecular forces between the molecules of the ice
  - As the forces are overcome, the solid water becomes liquid

Condensation & solidification

- Changes of state depend on whether energy is being transferred **to** or **away from** the system
  - Heating is when energy is transferred **to** the system and the kinetic energy of the molecules increases (red arrows to the right)
  - Cooling is when energy is transferred **away from** the system (or dissipated to the surroundings) and the kinetic energy of the molecules decreases (blue arrows to the left)

Condensation

- When a gas **cools**, energy is transferred away from the system and kinetic energy decreases until the temperature reaches **boiling point**
  - At boiling point, energy transferred away from the system reduces its potential energy
  - The particles no longer have enough energy to overcome the **intermolecular forces** of attraction
  - They only have enough energy to **flow** over one another
- The gas has condensed; it is now a liquid
  - As the energy has been transferred away from the **potential** store of the particles, the energy in the **kinetic** store is unchanged, so **temperature remains constant** through this process

Solidification

- When a liquid **cools**, energy is transferred away from the system and kinetic energy decreases until the temperature reaches **melting point**
  - At melting point, energy transferred away from the system reduces its potential energy
  - The particles no longer have enough energy to overcome the **intermolecular forces** of attraction
  - They are now low enough in energy to be bound to each other and can only vibrate around a fixed point
- The liquid has solidified; it is now a solid
  - As the energy has been transferred away from the **potential** store of the particles, the energy in the **kinetic** store is unchanged, so **temperature remains constant** through this process

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